

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	

Code of Conduct

I Vasanthan Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: "**Safari download for Windows**" and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	4	4	3	3	18	92
2	5	4	3	4	4	20	
3	4	5	4	3	4	20	
4	4	4	5	4	3	20	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1 — Frame Semantics in Banking Customer-Service Systems

1. Semantic Frame Analysis

Query	Semantic Frame	Action	Participants
Transfer ₹20,000 from my savings account to my son's account	TRANSFER(SourceAccount, Amount, DestinationAccount, Transfer Customer)	Transfer	Source = savings account, Amount = ₹20,000, Destination = son's account, Customer = user
Block my debit card immediately	BLOCK(DebitCard, Customer)	Block	Object = debit card, Customer = user
Send my transaction statement to my email	SEND(TransactionStatement, Email, Customer)	Send	Object = transaction statement, Destination = email, Customer = user
Increase my daily withdrawal limit	MODIFY(WithdrawalLimit, Customer)	Modify	Object = withdrawal limit, Customer = user

The frames correctly connect each **action** with the entities/participants involved.

2. Incorrect Semantic Interpretation

B4 is incorrect.

Actual intent: Increase withdrawal limit

System interpretation: Decrease withdrawal limit

The action “**increase**” is opposite to “**decrease**”, so the system has assigned the wrong semantic meaning.

3. Effect of Incorrect Semantic Roles

Incorrect semantic-role identification can cause the banking system to perform the **wrong operation**.

For B4:

User: "Increase my daily withdrawal limit."

Correct:

MODIFY(WithdrawalLimit, Customer)

→ Increase

Incorrect:

MODIFY(WithdrawalLimit, Customer)

→ Decrease

This could reduce the customer's withdrawal limit instead of increasing it, resulting in an incorrect banking operation.

4. Recommended Frame-Based Approach

Use a **frame-based semantic analysis pipeline**:

User Query

↓

Intent / Action Identification

↓

Semantic Role Identification

↓

Frame Construction

↓

Validate Participants & Values

↓

Banking Operation

For example:

"Transfer ₹20,000 from savings to son's account"

TRANSFER

/ | \

Source Amount Destination

Savings ₹20,000 Son's account

The system should validate the **action and all associated roles** before executing the banking operation. This reduces semantic interpretation errors and improves reliability.

Conclusion

Frame-based semantic analysis provides a structured representation of **actions and their participants**. Correctly identifying these roles is essential in banking systems because a semantic error can directly result in the wrong financial operation.

Q2 — First-Order Predicate Calculus for Smart Agriculture

1. Represent the Field Observations

Using predicates from the given data:

F1

Dry(F1)

IrrigationAvailable(F1)

Rice(F1)

F2

Wet(F2)

IrrigationAvailable(F2)

Wheat(F2)

F3

Dry(F3)

IrrigationUnavailable(F3)

Maize(F3)

F4

Wet(F4)

IrrigationUnavailable(F4)

Rice(F4)

The field conditions are given directly in the assessment table.

2. Determine Which Fields Require Irrigation

Given rules:

$\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$

$\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$

$\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$

$\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$

$\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

F1

$\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1})$

$\rightarrow \text{NeedsWater}(\text{F1})$

$\text{NeedsWater}(\text{F1})$

$\rightarrow \text{Irrigate}(\text{F1})$

Therefore:

F1 requires irrigation.

Since F1 is rice:

$\text{Rice}(\text{F1}) \wedge \text{NeedsWater}(\text{F1})$

$\rightarrow \text{PriorityIrrigation}(\text{F1})$

So F1 also has **priority irrigation**.

F2

$\text{Wet}(\text{F2})$

$\rightarrow \neg \text{NeedsWater}(\text{F2})$

Therefore:

F2 does not require irrigation.

F3

$\text{Dry}(\text{F3})$

but:

$\text{IrrigationUnavailable}(\text{F3})$

$\rightarrow \neg \text{Irrigate}(\text{F3})$

Therefore:

F3 needs water but cannot be irrigated.

F4

Wet(F4)

→ $\neg$ NeedsWater(F4)

Therefore:

F4 does not require irrigation.

Final Result

	Field Needs Water	Can Irrigate	Decision
F1	Yes	Yes	Irrigate + Priority
F2	No	Yes	No irrigation
F3	Dry/needs water	No	Cannot irrigate
F4	No	No	No irrigation

3. F1 vs F3

Yes, **F1 and F3 must be treated differently.**

F1 → Dry + Irrigation Available

→ NeedsWater

→ Irrigate

→ PriorityIrrigation

F3 → Dry + Irrigation Unavailable

→ Cannot Irrigate

So although both fields are **dry**, only F1 can be automatically irrigated.

4. Is Predicate Logic Alone Sufficient?

No.

Predicate logic works well when sensor information is **clear and certain**, but real agricultural systems can have:

- incomplete sensor readings
- uncertain soil conditions
- conflicting sensor values

- changing environmental conditions

For example, if the soil sensor cannot determine whether a field is dry or wet, a strict rule such as:
 $\text{Dry}(x) \rightarrow \text{NeedsWater}(x)$

cannot reliably make a decision.

Therefore, predicate logic should be combined with **uncertainty handling and real-time sensor analysis** for reliable agricultural decision support.

Conclusion

F1 is the only field that should currently be **automatically irrigated**, with priority because it is rice. F3 requires water but cannot be irrigated because irrigation is unavailable. This demonstrates both the usefulness and limitations of predicate-based reasoning.

Q3 — Word Sense Disambiguation in a Travel Recommendation System

1. Determine the Intended Sense

Query	Intended Sense	Evidence
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages
Safari booking	Wildlife tour	User selected wildlife resort packages
Java vacation	Java island	User viewed hotels in Bali and Java
Cruise package	Ship journey	User viewed Mediterranean ship packages
Safari download for Windows	Web browser	User clicked a browser download page

The first four travel-related interactions strongly indicate the intended travel sense, while the last query clearly indicates the **browser** sense from “download for Windows” and the clicked browser page.

2. Lexical and Contextual Cues

Ambiguous Word Travel Cue		Other-Sense Cue
Amazon	tour, trekking	company
Safari	booking, wildlife resort	download, Windows
Java	vacation, hotels	programming

Ambiguous Word Travel Cue

Other-Sense Cue

Cruise package, Mediterranean software/project

The surrounding words provide strong contextual clues. User interaction history provides an additional signal for selecting the correct sense.

3. Why the Same Word Has Different Meanings

A word can have multiple **lexical senses**, and its intended meaning depends on the context and the user's goal.

For example:

Safari booking

→ Wildlife tour

Safari download for Windows

→ Web browser

Although both contain **Safari**, the surrounding context and user behaviour indicate completely different meanings. Therefore, a WSD system should not rely only on the ambiguous word itself.

4. Industrial-Scale WSD Strategy

A robust travel recommendation system can combine four sources:

Search Query



Context Extraction



User History ———┐



WSD Model



Embeddings ———┐



Click Behaviour ———┐



Correct Word Sense



Travel Recommendation

Approach

1. **Query context:** Use surrounding words to identify the likely sense.
2. **User history:** Use previous searches and interactions to personalize interpretation.
3. **Embeddings:** Represent the query and possible senses as vectors and compare their contextual similarity.
4. **Click behaviour:** Use clicked pages/packages as feedback to confirm or update the selected sense.
5. **Continuous learning:** Use user interactions to improve future disambiguation.

This combination is more reliable than using a single clue because the assessment specifically recommends using **query context, user history, embeddings, and click behaviour** for industrial-scale WSD.

Conclusion

Context and user behaviour are essential for WSD. The system should dynamically determine the intended sense instead of assigning one fixed meaning to an ambiguous word.

Q4 — Syntax-Driven Semantic Analysis in Legal Document Processing

1. Effect of Syntactic Structure on Semantic Roles

Syntactic structure helps identify **who performs the action, what is affected, and who receives it**.

Active voice

The supplier delivered the equipment to the buyer.

- Supplier → **Agent**
- Equipment → **Object**
- Buyer → **Recipient**

Here, the grammatical subject is also the semantic agent.

Another active sentence

The buyer received the equipment from the supplier.

- Buyer → **Agent**
- Equipment → **Object**
- Supplier → **Source**

The prepositional phrase “**from the supplier**” identifies the source of the action.

Passive voice

The contract was terminated by the company.

Although “**contract**” is the grammatical subject, it is not the agent.

- Contract → **Object/Patient**
- Company → **Agent**

This shows why semantic roles cannot be assigned from grammatical subject alone.

2. Incorrect Semantic-Role Assignments

Incorrect 1

The buyer received the equipment from the supplier.

Given interpretation:

Buyer = Agent

Equipment = Object

Supplier = Recipient

Error: Supplier = Recipient

Correct:

Buyer = Agent

Equipment = Object

Supplier = Source

Incorrect 2

The contract was terminated by the company.

Given:

Contract = Agent

Company = Object

Incorrect.

Correct:

Contract = Object/Patient

Company = Agent

The assessment's provided interpretations contain these role-reversal errors.

3. Effect of Confusing Subject with Agent

A grammatical subject is **not always the semantic agent**, especially in passive constructions.

For example:

The contract was terminated by the company.

If the system assumes:

Subject → Agent

it would incorrectly conclude:

Contract → performed the termination

instead of:

Company → performed the termination

Contract → received the action

In legal NLP, this can lead to **incorrect extraction of responsibility, actions, and entities**, potentially changing the meaning of a contract or legal document.

4. Recommended Architecture

Legal Document



Sentence Segmentation



Syntactic Parser



Dependency / Constituency Analysis



Voice Detection



Semantic Role Labeling



Entity + Role Representation



Legal Information Extraction

The architecture should specifically handle:

- **Active voice**
- **Passive voice**
- **Prepositional phrases**
- **Long-distance dependencies**

For example:

"The contract was terminated by the company."

terminated	
/	\
contract	company
Object	Agent

Thus, combining syntactic parsing with semantic-role analysis prevents the system from simply equating **grammatical subject** = **semantic agent**.

Conclusion

Syntax provides the structural information needed for semantic interpretation, but **semantic roles must be determined from the complete syntactic structure**, including voice and prepositional relationships. This is especially important in legal NLP where incorrect role assignment can change the extracted meaning.